

- 1** Using an ohm-meter, a student measures the resistance of two resistors R_1 and R_2 separately with their associated uncertainties shown as follows:

$$R_1 = 200 \pm 5 \, \Omega$$

$$R_2 = 800 \pm 5 \, \Omega$$

The two resistors R_1 and R_2 are then connected in parallel. He calculated the effective resistance, R to be $160 \, \Omega$. What is the uncertainty of the effective resistance?

A $2.5 \, \Omega$

B $3.4 \, \Omega$

C $5 \, \Omega$

D $10 \, \Omega$